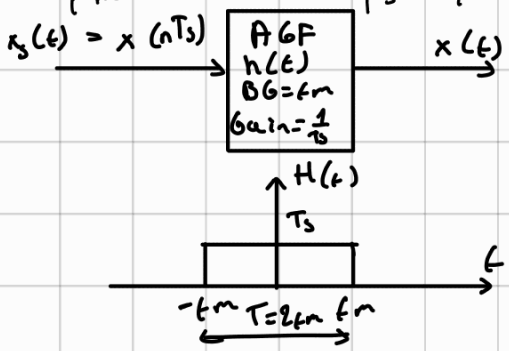
Fouad Aladhami
210 10200 2508

1)The signal $x(t)$ whose bandwidth is 50 Hz is ideally sampled at the Nyquist rate. Sample values are given below:

$$x(nT_s)=\begin{cases}-1 & -4\le n<0\\ 1 & 0<n\le 4\\ 0 & Diger\end{cases}$$

$x(0,005)=?$

$$\text{sinc}(t)=\frac{\sin(\pi t)}{\pi t}$$



Nyquist rate $\Rightarrow f_s = 100\,Hz$ and $T_s=\frac{1}{f_s}=0,001\ s$

$$H(f)=T_s\,\Pi\left(\frac{f}{2f_m}\right)\xrightarrow{F^{-1}} h(t)=2f_mT_s\sin(2f_mt)$$

$$x(t)=h(t)*x_s(t)=h(t)\circledast\sum_{n=-\infty}^{\infty}x(nT_s)\,\delta(t-nT_s)$$

$$x(t)=\sum_{n=-\infty}^{\infty}x(nT_s)\,h(t-nT_s)=2f_mT_s\sum_{n=-\infty}^{\infty}x(nT_s)\,\text{sinc}[2f_m(t-nT_s)]$$

$$x(t)=\sum_{n=-\infty}^{\infty}x(nT_s)\,\text{sinc}(100t-n)$$

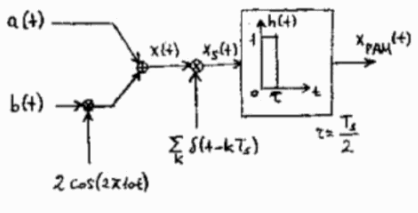
$$x(0,005)=\sum_{n=-4}^{4}x(nT_s)\,\text{sinc}(0,5-n)$$

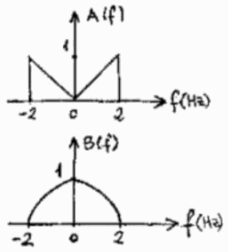
$$=-\sum_{n=-4}^{-1}\text{sinc}(0,5-n)+\sum_{n=1}^{4}\text{sinc}(0,5-n)=-[\text{sinc}(1,5)+\text{sinc}(2,5)+\text{sinc}(3,5)+\text{sinc}(4,5)]$$

$$+[\text{sinc}(-0,5)+\text{sinc}(-1,5)+\text{sinc}(-2,5)+\text{sinc}(-3,5)]=-[\text{sinc}(4,5)-\text{sinc}(0,5)]$$

$$=-\left[\frac{\sin(4,5\pi)}{(4,5\pi)}-\frac{\sin(0,5\pi)}{(0,5\pi)}\right]=0,566$$

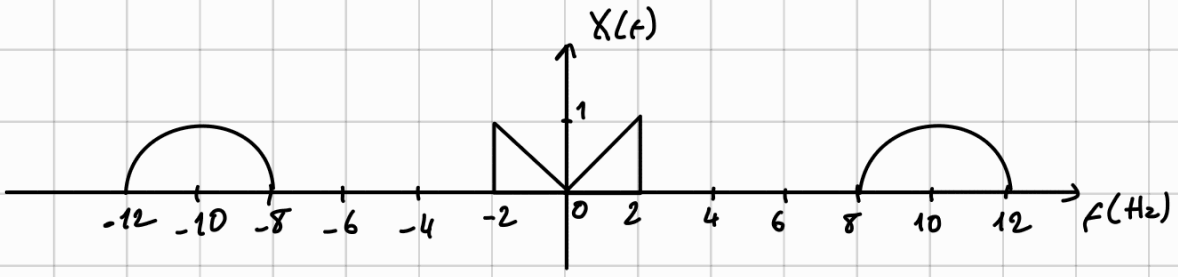
2)The signals $a(t)$ and $b(t)$ are applied to the input of the system given in the figure below. The signal $x(t)=a(t)+2b(t)Cos(2\pi 10t)$ is ideally sampled and samples are passed through the hold filter. The PAM signal $x_{PAM}(t)$ is obtained at the output of the hold filter. a) Sketch the spectrum of $x(t)$. b) Find the minimum sampling frequency. Skecth the spectra of $x_s(t)$ ve $x_{PAM}(t)$ for this frequency. c) Sketch the receiver to obtain $a(t)$ ve $b(t)$.





a) $x(t)=a(t)+2b(t)\cos(2\pi\,10t)$

$$X(f)=A(f)+B(f-10)+B(f+10)$$



b) $f_m=12\,Hz\Rightarrow f_s=2f_m=24\,Hz$ Nyquist Rate

$$x_s(t)=x(t)\sum_k\delta(t-kT_s)=\sum_k x(kT_s)\,\delta(t-kT_s)$$

$$X_s(f)=X(f)\circledast f_s\sum_k\delta(f-kf_0)=f_s\sum_k X(f-kf_s)=24\sum_k X(f-24k)$$

F ↳ $x_{PAM}(t) = x_s(t) \circledast h(t)$

$X_{PAM}(f) = X_s(f)\, H(f)$

$= \left(24 \sum_k X(f-24k)\right)\left(\frac{1}{48} \text{sinc}\left(\frac{1}{48} f\right) e^{-j2\pi f \frac{1}{96}}\right)$

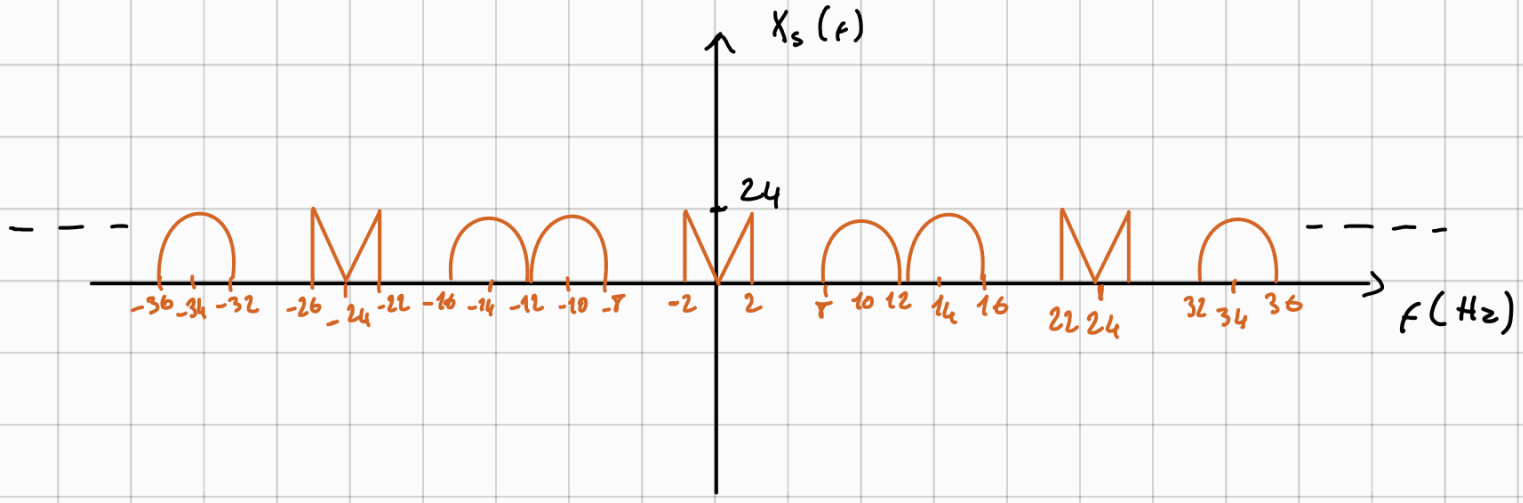
$h(t) = \sqcap\left(\frac{t-z/2}{z}\right)$

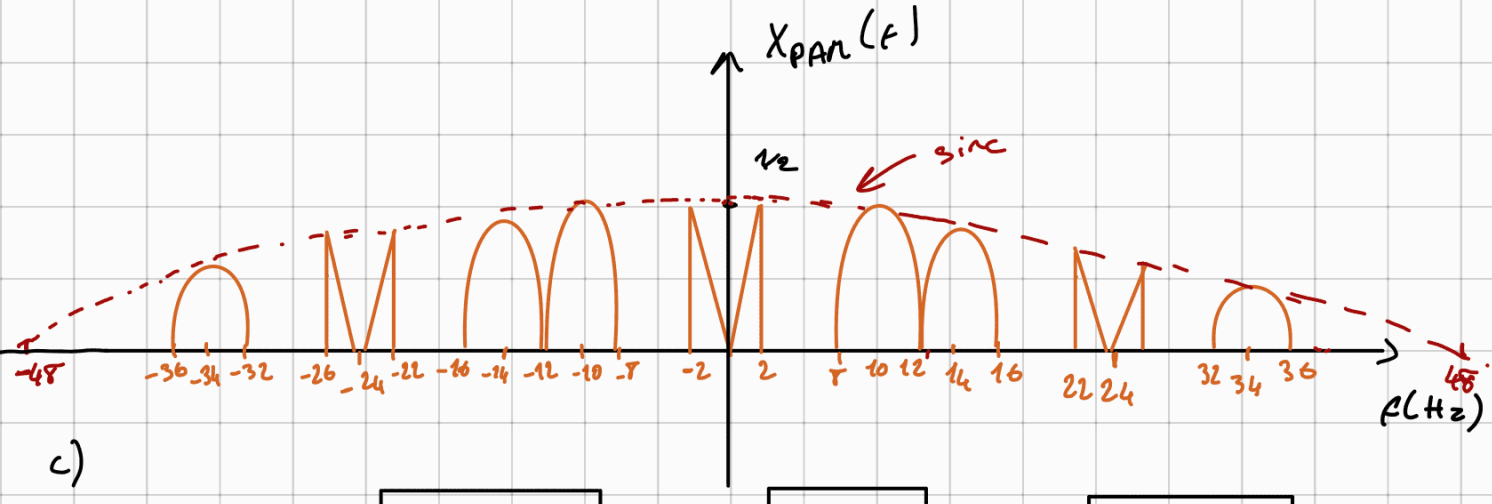
↳ $H(f) = z\,\text{sinc}(fz)\, e^{-j2\pi f \frac{z}{2}}$

when $z = \frac{T_s}{2} = \frac{1}{48}$

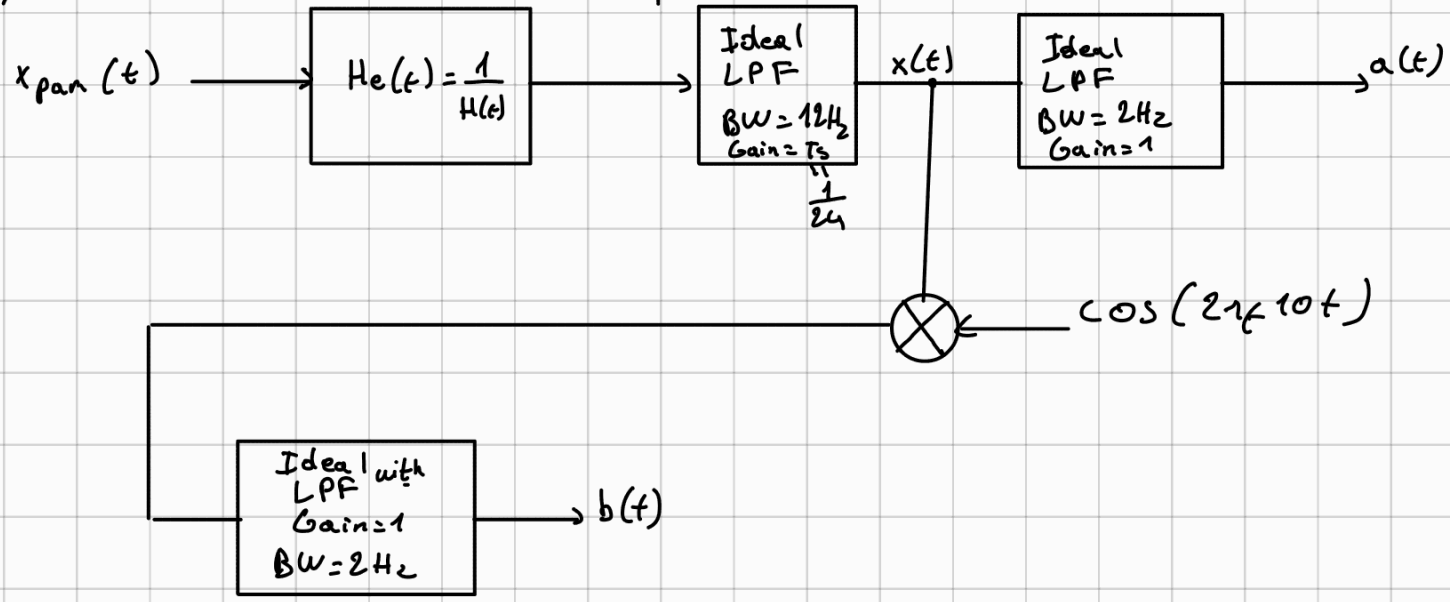
$H(f) = \frac{1}{48} \text{sinc}\left(\frac{1}{48} f\right) e^{-j2\pi f \frac{1}{96}}$

$X_{PAM}(f) = \frac{1}{2} \text{sinc}\left(\frac{1}{48} f\right) e^{-j2\pi f \frac{1}{96}} \sum_k X(f-24k)$

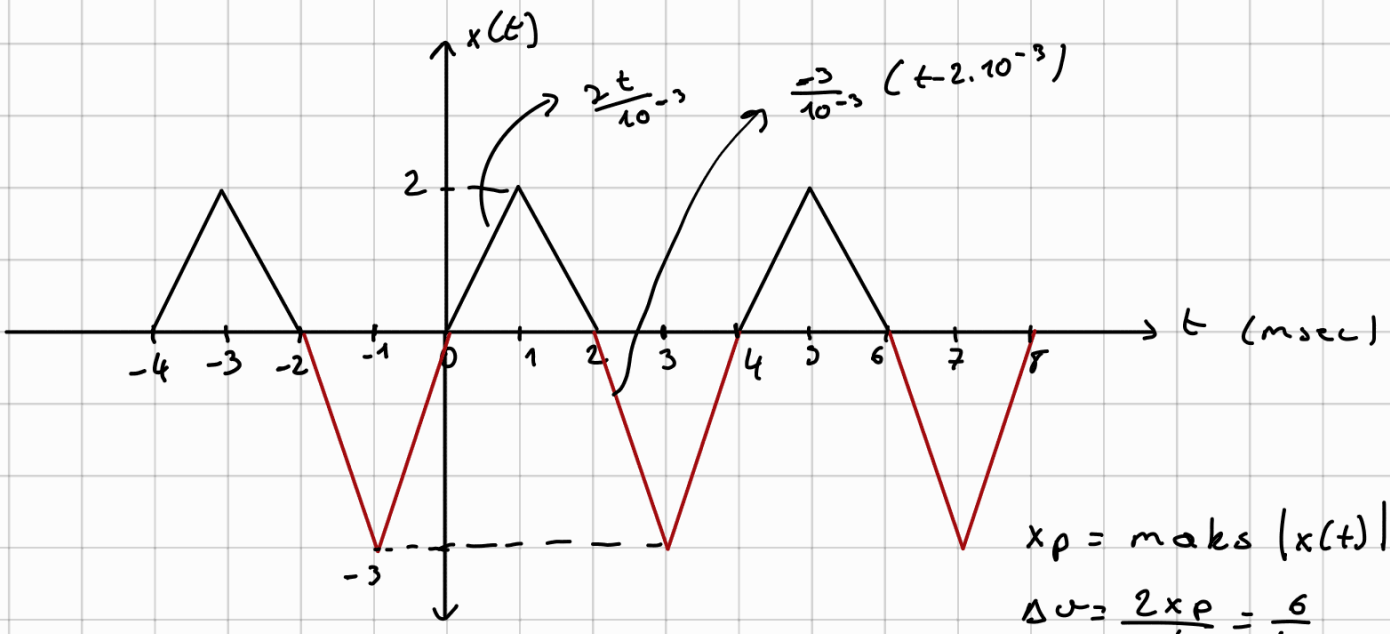




c)



3)The signal $x(t)$ whose one period is expressed as $2\Lambda(t-1)-3\Lambda(t-3)$ is applied to uniform PCM modulator. If SQNR is required to be at least 40 dB, find the transmission rate and minimum required channel bandwidth at the output of PCM modulator. ($\Lambda(t)$ is triangular pulse signal. Unit of t is msec, the delays 1 and 3 are in msec.)



$x_p = \max |x(t)| = 3$

$\Delta v = \frac{2x_p}{L} = \frac{6}{L}$

$x_q(t) = x(t) + q(t)$ → Quantization Noise

$x_q(t)$: Quantized signal; $x(t)$: Analog noise

q is uniform in $\left[-\frac{\Delta v}{2}, \frac{\Delta v}{2}\right]$

Average quantization Noise power: $N_q = E[q^2]$

$$N_q = \frac{\left[\frac{\Delta v}{2} - \left(-\frac{\Delta v}{2}\right)\right]^2}{12} = \frac{\Delta v^2}{12}$$

$$\Delta v = \frac{2x_p}{L} \Rightarrow N_q = \frac{\Delta v^2}{12} = \frac{x_p^2}{3L^2}$$

$$SNR = \frac{S_o}{N_o} = \frac{P_x}{x_p^2} \cdot 3L^2$$

$$P_x = \frac{1}{T}\int_0^4 |x(t)|^2 dt$$

$$= \frac{1}{4\cdot 10^{-3}}\left[2\int_0^{10^{-3}} \left|\frac{2t}{10^{-3}}\right|^2 dt + 2\int_{2\cdot 10^{-3}}^{3\cdot 10^{-3}} \left|\frac{-3t - 2\cdot 10^{-3}}{10^{-3}}\right|^2\right]$$

$$= \frac{1}{4\cdot 10^{-3}}\left[2\int_0^{10^{-3}} \left|\frac{2t}{10^{-3}}\right|^2 dt + 2\int_0^{10^{-3}} \left|\frac{-3t}{10^{-3}}\right|^2 dt\right]$$

$$= \frac{1}{4\cdot 10^{-3}}\left[2\int_0^{10^{-3}} \frac{4t^2}{10^{-6}} dt + 2\int_0^{10^{-3}} \frac{9t^2}{10^{-6}} dt\right]$$

$$= \frac{1}{4\cdot 10^{-3}} \left[\frac{8}{10^{-6}} \frac{t^3}{3} + \frac{18}{10^{-6}} \frac{t^3}{3} \right]_{t=10^{-3}}$$

$$= \frac{1}{4\cdot} \left[\frac{26}{3} 10^{-3} \cdot 10^{3} \right]$$

$$= \frac{13}{6} \quad W \approx 2,167 \quad W$$

- $SNR_{db} = 40$ db at minimum

$40 = 10 \log(SNR)$

$\Rightarrow \log(SNR) = 4$

$SNR = 10^4$ //

$$SNR = \frac{S_0}{N_0} = \frac{P_x}{\frac{x^2 \rho}{3L^2}} = 10^4$$

$$2,167 \cdot 10^{-4} = \frac{\left(\frac{13}{6}\right)}{10^4} = N_0$$

$$\frac{x^2 \rho}{3L^2} = 2,167 \cdot 10^{-4}$$

$$L^2 = \frac{x^2 \rho = 3}{3 (2,167 \cdot 10^{-4})} = 13844,024$$

$\Rightarrow L = 117,7$

$L = 2^n$ olmalıdır

buna en yakın olan 2^7 olan 128'dir

$\Rightarrow n = 7$

$\Rightarrow R_b = n f_{s_{min}} = 7 \cdot 2k = 14k \frac{bit}{s}$

$\Rightarrow B_k = \frac{R_b}{2} = 7$ khz

$A \sqcap\!\!\sqcap \left(\frac{t}{2}\right) \longleftrightarrow A 2 \sin c^2(f 2)$

$\frac{1}{2}$ 'da x eksenine keser

yaklaşık olarak bant genişliği

$BG = \frac{1}{10^{-3} \cdot 2} = 1$khz (%90 power)

$\Rightarrow f_m = \frac{1}{2} = 10^3 = 1$khz

$\Rightarrow f_s \geq 2 f_m \Rightarrow f_s \geq 2$khz

$f_{s_{min}} = 2$khz

4)

In a single-integration DM system, the voice signal is sampled at a rate of 64 kHz, similar to PCM. The maximum signal amplitude is normalized as $A_{max} = 1$.

(a) Determine the minimum value of the step size Δ to avoid slope overload.

(b) Determine the granular noise power N_o if the voice signal bandwidth is 3.4 kHz.

(c) Assuming that the voice signal is sinusoidal, determine S_o and the SNR.

(d) Assuming that the voice signal amplitude is uniformly distributed in the range $(-1, 1)$, determine S_o and the SNR.

(e) Determine the minimum transmission bandwidth.

Hint: Voice signal: $x(t) = A_{max}\, Sin(2\pi 800t)$

$$\max |\dot{x}(t)| \leq \frac{\Delta}{T_s} \qquad f_s = 64\,khz$$

$$\dot{x}(t) = 1600\pi\, A_{max} \sin(2\pi 800t)$$

$$1600\pi A_{max} \cdot T_s \leq \Delta$$

$$\frac{1600\pi}{64\,000} \leq \Delta \quad \Rightarrow \quad \Delta \geq \frac{1}{40}\pi$$

b) $$N_q = \frac{\Delta^2 B}{3 f_s} = \frac{\left(\frac{1}{40}\pi\right)^2 \cdot 3{,}4\,k}{3 \cdot 64\,k} = 109{,}2\,\mu W$$

c) $$S_o = P_x = \frac{A^2_{max}}{2} = \frac{1}{2} \text{ watt}$$

because it is sinusoidal

$$SNR = \frac{S_o}{N_q} = 4577{,}3$$

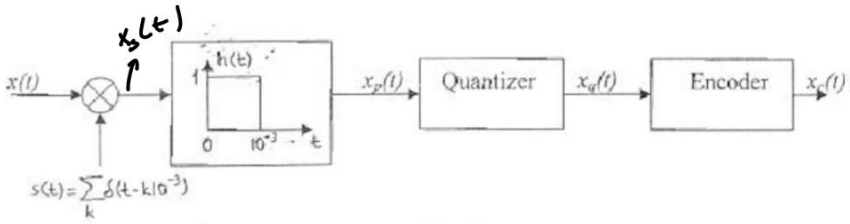
d) Because the $x(t)$ signal is uniformly distributed between -1 and 1

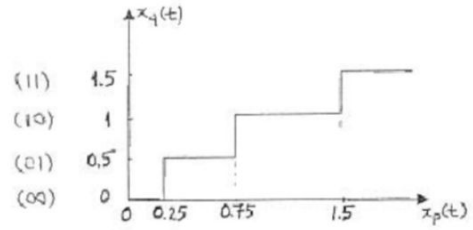
$$S_o = \frac{1}{12}(b-a)^2 = \frac{2^2}{12} = \frac{4}{12} = \frac{1}{3} \quad \text{so} \quad SNR = \frac{S_o}{N_q} = \frac{1/3}{109{,}2\mu} = 3051{,}5$$

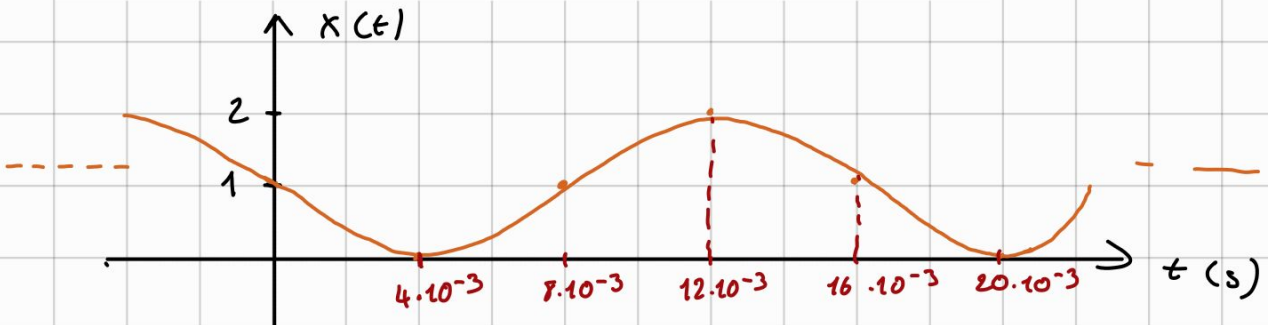
e) $$BG = \frac{n f_s}{2} = \frac{1 \cdot 64\,khz}{2} = 32\,khz$$

1 bit for delta modulation

5)The signal $x(t) = 1 - Sin(1000\pi t/8)$ is applied to the input of the system given in the figure below. Encoder maps the quantization levels (0), (0,5), (1), (1,5) to the code words 00, 01, 10, 11, respectively. The bits "0" ve "1" are transmitted through the channel with a rectangular pulse of width 0.5 msec and amplitudes 0 Volt and 1 Volt, respectively. a) Sketch $x(t)$, $x_p(t)$, $x_q(t)$ and $x_c(t)$. b) Find the transmission rate and minimum required channel bandwidth for the PCM signal.

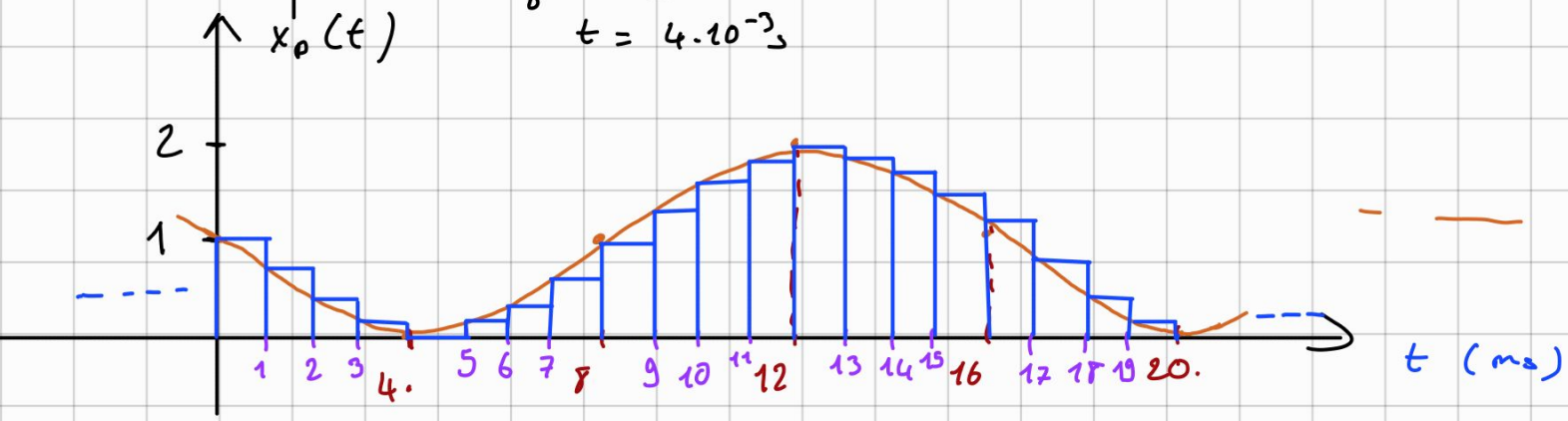


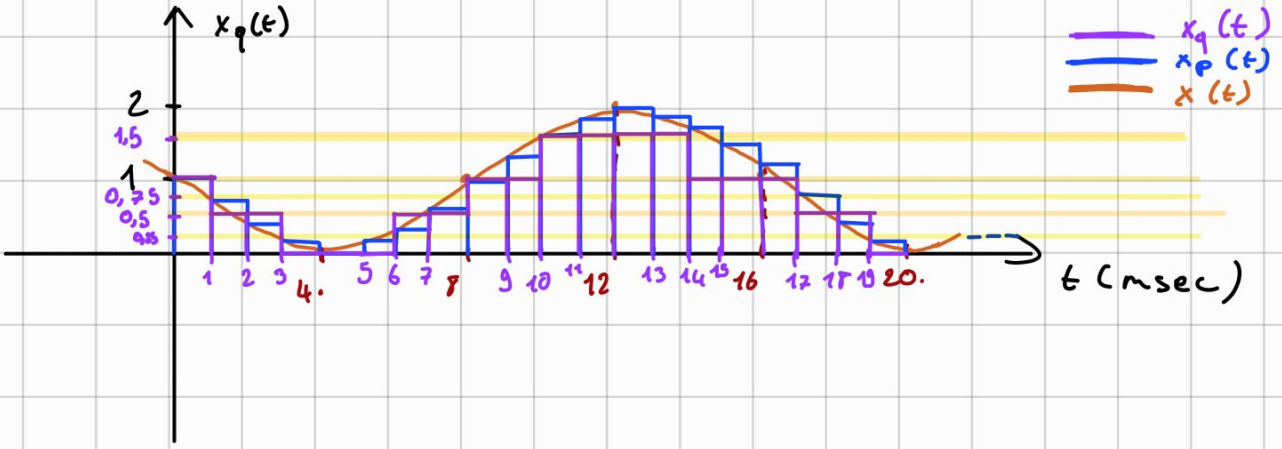


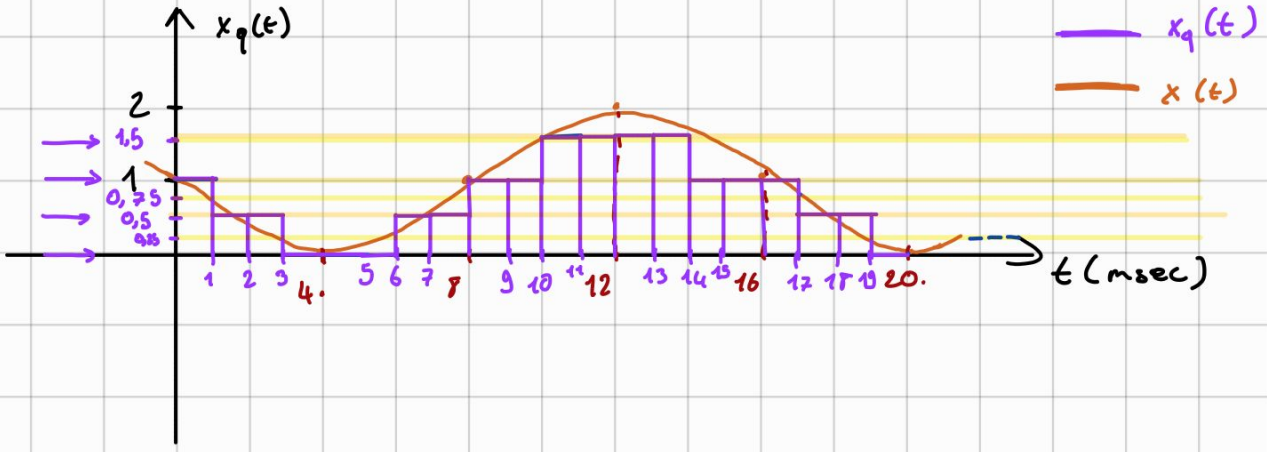


$\frac{1000\pi t}{8} = \frac{\pi}{2}$

$t = 4.10^{-3}$



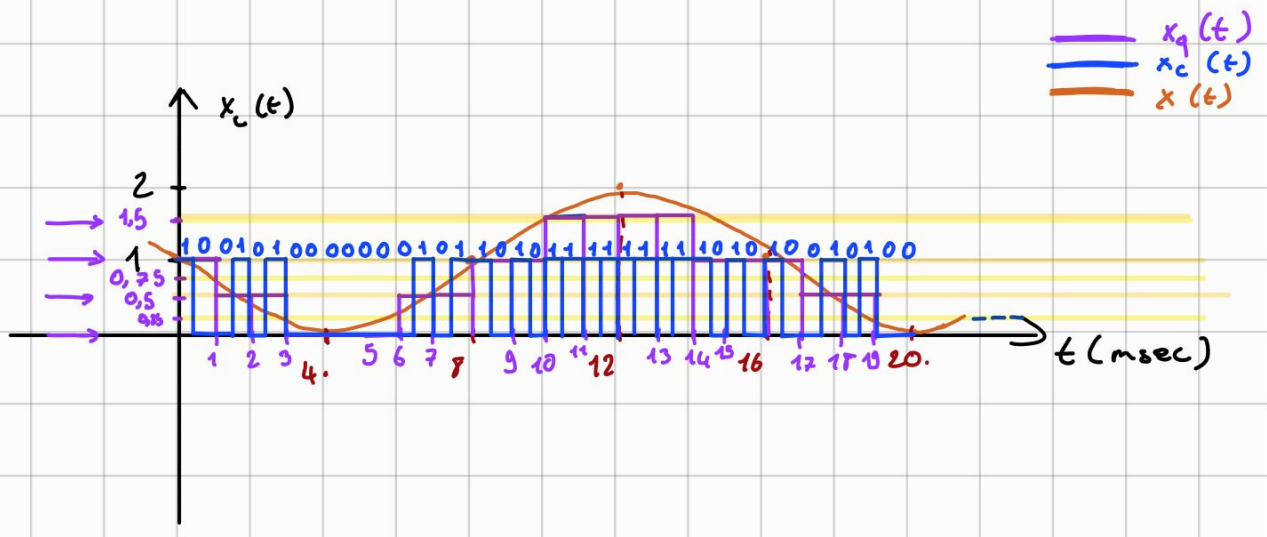




0 V → 00
0,5V → 01
1 V → 10
1,5V → 11

0 → 0V
1 → 1V

(0,5 msec rect.)



İletim hız $R_b = n f_s \frac{bit}{sn}$

$f_s = \frac{1}{T_s} = \frac{1}{10^{-3}} = 1 khz$

$R_b = n f_s = 2k \text{ bit/s}$ (Transmission rate)

Min required BG $B_k = \frac{R_b}{2} = 1 khz$